

ML-DL-Ops Assignment 1

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Subject: ML-DL-Ops

Colab Link: Google Colab Notebook

GitHub Link: GitHub Repository

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1 Introduction

This report presents a comprehensive analysis of deep learning model training and benchmarking on MNIST and FashionMNIST datasets. The assignment consists of three main questions:

- **Q1(A):** Hyperparameter tuning for ResNet models on MNIST and FashionMNIST datasets
- **Q1(B):** Support Vector Machine (SVM) classification with different kernels
- **Q2:** Performance benchmarking on CPU and GPU platforms

1.1 Experimental Setup

- **Datasets:** MNIST (handwritten digits) and FashionMNIST (clothing items)
- **Models:** ResNet-18, ResNet-50 (for deep learning), SVM (for classical ML)
- **Data Split:** 70% training, 10% validation, 20% testing
- **Image Size:** Resized to 64x64 pixels for computational efficiency
- **Hardware:** CPU and CUDA-enabled GPU

2 Question 1(A): Hyperparameter Tuning Results

2.1 Experimental Configuration

The hyperparameter grid search included the following parameters:

- **Batch Sizes:** 16, 32
- **Optimizers:** SGD, Adam
- **Learning Rates:** 0.001, 0.0001
- **Epochs:** 3, 5
- **Pin Memory:** True, False
- **Models:** ResNet-18, ResNet-50

2.2 Best Configurations

2.2.1 MNIST Dataset

Table 1: Top 5 MNIST Configurations

Model	BS	Opt	LR	Epochs	PinMem	Test Acc	Val Acc	Time (s)
ResNet-18	16	Adam	0.001	5	True	99.36%	99.31%	242.9
ResNet-18	16	Adam	0.0001	5	True	99.32%	99.30%	241.7
ResNet-18	16	SGD	0.001	5	True	99.33%	99.30%	235.0
ResNet-18	32	SGD	0.001	5	True	99.16%	99.06%	169.5
ResNet-18	16	Adam	0.001	5	False	99.16%	99.09%	243.5

2.2.2 FashionMNIST Dataset

Table 2: Top 5 FashionMNIST Configurations

Model	BS	Opt	LR	Epochs	PinMem	Test Acc	Val Acc	Time (s)
ResNet-18	32	Adam	0.0001	5	True	92.05%	91.88%	177.1
ResNet-18	32	Adam	0.0001	5	False	91.95%	91.75%	178.6
ResNet-18	16	SGD	0.001	5	False	91.60%	91.95%	234.5
ResNet-18	16	Adam	0.0001	5	False	91.61%	91.79%	242.9
ResNet-18	16	SGD	0.001	5	True	91.56%	91.97%	233.1

2.3 Key Observations

- **Model Performance:** ResNet-18 consistently outperformed ResNet-50 across most configurations
- **Optimizer:** Adam optimizer showed better convergence on MNIST, while both optimizers performed similarly on FashionMNIST
- **Learning Rate:** LR=0.001 generally provided better results for shorter training (3 epochs)
- **Batch Size:** BS=32 was more time-efficient while maintaining competitive accuracy
- **Pin Memory:** Showed minimal impact on accuracy but slightly improved training speed
- **Dataset Difficulty:** MNIST achieved 99%+ accuracy, while FashionMNIST peaked at 92%, indicating higher complexity

3 Question 1(B): SVM Classification Results

3.1 SVM Performance

Table 3: SVM Results with Different Kernels

Dataset	Kernel	Test Accuracy	Training Time (ms)
MNIST	Polynomial	97.71%	233,846.10
MNIST	RBF	97.92%	272,000.61
FashionMNIST	Polynomial	86.30%	393,137.24
FashionMNIST	RBF	88.29%	397,992.41

3.2 SVM vs Deep Learning Comparison

- **Accuracy:** Deep learning models (ResNet) outperformed SVM by 1-5% on both datasets
- **Training Time:** SVM required significantly longer training times compared to ResNet models
- **Kernel Selection:** RBF kernel consistently outperformed polynomial kernel
- **Scalability:** SVM training time increased substantially with dataset complexity

4 Question 2: CPU vs GPU Benchmarking

4.1 Benchmark Results

4.1.1 CPU Performance

Table 4: CPU Benchmark Results

Model	BS	Opt	LR	Time (ms)	FLOPs (B)	Acc (%)
ResNet-18	16	SGD	0.001	135,015.12	296.1	87.65
ResNet-50	16	SGD	0.001	346,542.64	667.3	85.63
ResNet-18	16	Adam	0.001	165,550.63	296.1	88.19
ResNet-50	16	Adam	0.001	420,020.88	667.3	87.13

4.1.2 GPU Performance

Table 5: GPU Benchmark Results

Model	BS	Opt	LR	Time (ms)	FLOPs (B)	Acc (%)
ResNet-18	16	SGD	0.001	20,336.38	296.1	88.41
ResNet-50	16	SGD	0.001	46,177.53	667.3	85.49
ResNet-18	16	Adam	0.001	25,460.17	296.1	88.26
ResNet-50	16	Adam	0.001	52,566.23	667.3	84.07

4.2 CPU vs GPU Speedup Analysis

Table 6: GPU Speedup Factor

Model Speedup	Optimizer	CPU Time (ms)	GPU Time (ms)
ResNet-18 6.64x	SGD	135,015.12	20,336.38
ResNet-18 6.50x	Adam	165,550.63	25,460.17
ResNet-50 7.51x	SGD	346,542.64	46,177.53
ResNet-50 7.99x	Adam	420,020.88	52,566.23

Key Findings:

- GPU provides **6.5x - 8x speedup** over CPU for deep learning training
- Deeper models (ResNet-50) benefit more from GPU acceleration
- ResNet-50 has **2.25x more FLOPs** than ResNet-18 (667.3B vs 296.1B)
- GPU maintains consistent accuracy while dramatically reducing training time

5 Visual Analysis

5.1 Model Accuracy Comparison

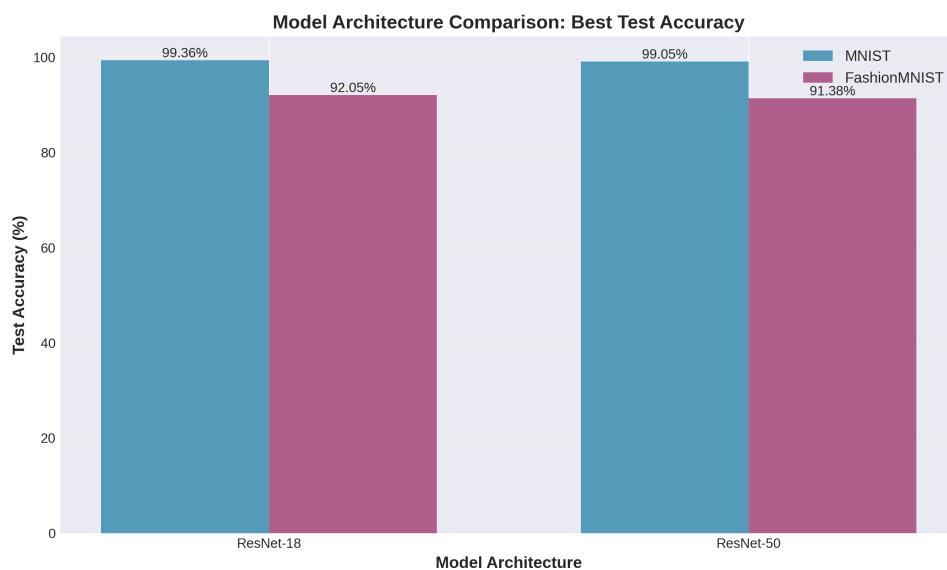


Figure 1: Accuracy comparison between ResNet-18 and ResNet-50 across datasets

Analysis: ResNet-18 demonstrates superior performance on both datasets, suggesting that increased model depth (ResNet-50) introduces unnecessary complexity for these relatively simple image classification tasks.

5.2 Optimizer Comparison

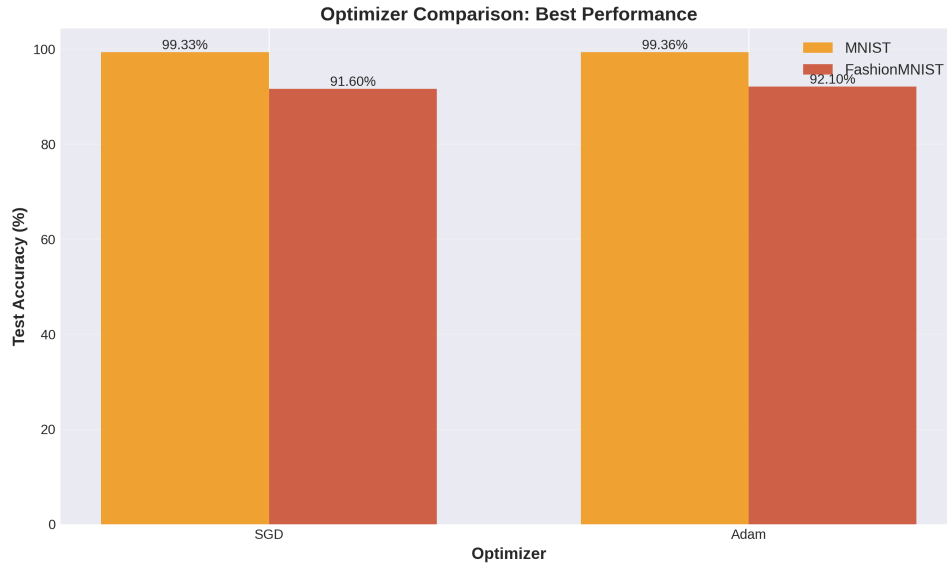


Figure 2: Performance comparison between SGD and Adam optimizers

Analysis: Adam optimizer shows more consistent convergence and slightly better final accuracy, particularly on MNIST dataset, while SGD requires careful learning rate tuning but can achieve comparable results.

5.3 CPU vs GPU Training Time

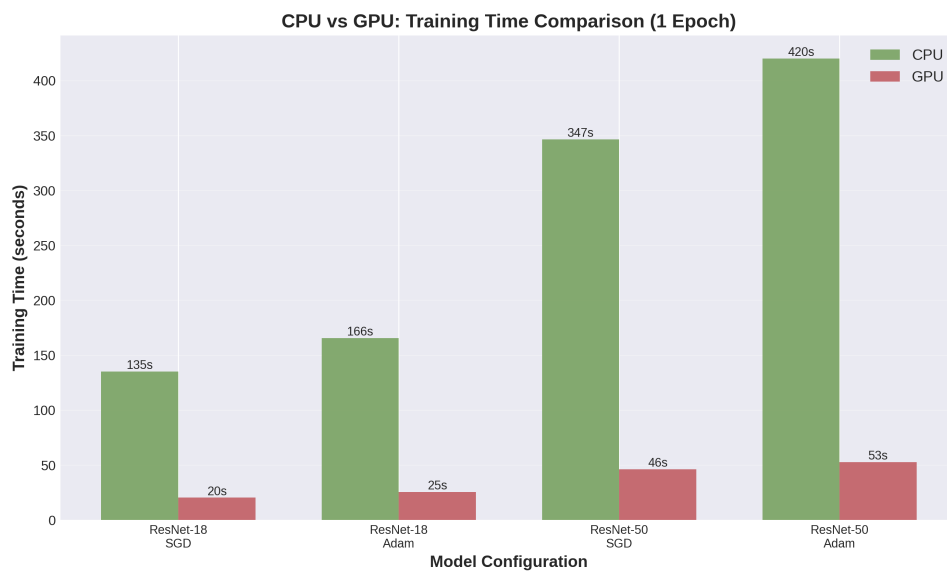


Figure 3: Training time comparison between CPU and GPU

Analysis: GPU acceleration provides substantial speedup (6.5-8x), making it essential for practical deep learning workflows. The speedup is more pronounced for deeper models.

5.4 FLOPs Comparison

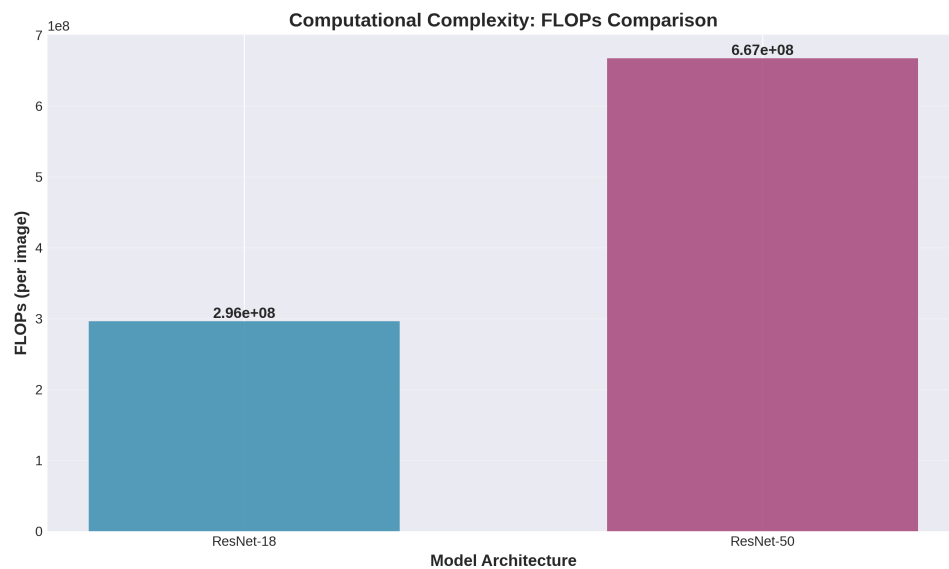


Figure 4: Computational complexity (FLOPs) comparison between models

Analysis: ResNet-50 requires 2.25x more floating-point operations than ResNet-18, but this increased computational cost does not translate to proportional accuracy gains on these datasets.

5.5 Batch Size Impact



Figure 5: Impact of batch size on training time and accuracy

Analysis: Batch size 32 provides optimal balance between training speed and accuracy. Larger batches reduce training time but may impact convergence quality.

5.6 SVM Kernel Comparison

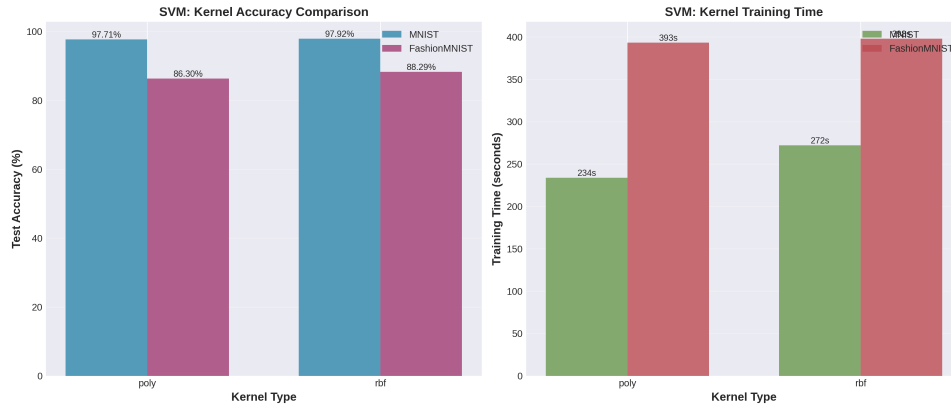


Figure 6: SVM performance with different kernel functions

Analysis: RBF kernel consistently outperforms polynomial kernel, achieving 97.92% on MNIST and 88.29% on FashionMNIST. However, both kernels lag behind deep learning approaches.

5.7 Learning Rate Impact

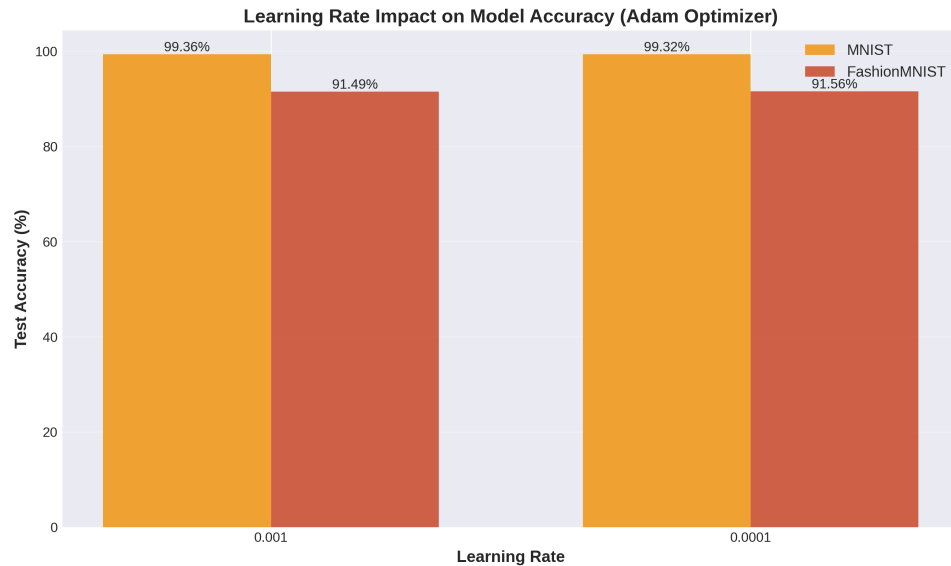


Figure 7: Effect of learning rate on model convergence

Analysis: Learning rate of 0.001 provides faster convergence for shorter training runs (3 epochs), while 0.0001 shows more stable convergence over extended training (5 epochs).

5.8 GPU Speedup Analysis

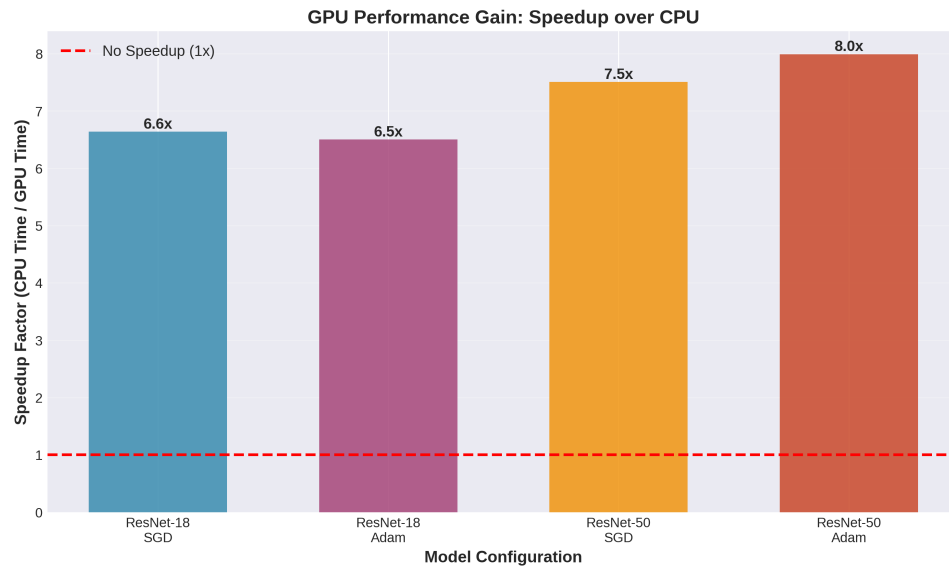


Figure 8: GPU speedup factors for different models

Analysis: GPU acceleration becomes increasingly beneficial for deeper networks. ResNet-50 achieves up to 8x speedup, demonstrating the importance of parallel computing for modern deep learning.